**Q1 - 24 January - Shift 1**

The value $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ is

(1) ${}^{45}C_{23}$

(2) ${}^{44}C_{23}$

(3) ${}^{45}C_{24}$

(4) ${}^{44}C_{22}$

*Space for your notes:*

**Q2 - 24 January - Shift 1**

Suppose $\sum_{r=0}^{2023} r^2 \, {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of $\alpha$ is ________

*Space for your notes:*

**Q3 - 24 January - Shift 2**

If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots\ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$, then $\alpha$ is equal to

(1) 30
(2) 60
(3) 15
(4) 10

*Space for your notes:*

**Q4 - 24 January - Shift 2**

Let the sum of the coefficients of the first three terms in the expansion of $\left(x-\frac{3}{x^2}\right)^n$, $x \neq 0$, $n \in N$, be 376. Then the coefficient of $x^4$ is ________.

*Space for your notes:*

**Q5 - 25 January - Shift 1**

If $a_r$ is the coefficient of $x^{10-r}$ in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to

(1) 4895
(2) 1210
(3) 5445
(4) 3025

*Space for your notes:*

**Q6 - 25 January - Shift 1**

The constant term in the expansion of

$\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is _____ .

*Space for your notes:*

**Q7 - 25 January - Shift 2**

$\sum_{k=0}^{6} {}^{51-k}C_3$ is equal to

(1) ${}^{51}C_4 - {}^{45}C_4$
(2) ${}^{51}C_3 - {}^{45}C_3$
(3) ${}^{52}C_4 - {}^{45}C_4$
(4) ${}^{52}C_3 - {}^{45}C_3$

*Space for your notes:*

**Q8 - 25 January - Shift 2**

The remainder when $(2023)^{2023}$ is divided by 35

*Space for your notes:*

**Q9 - 29 January - Shift 1**

If the co-efficient of $x^9$ in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ and the co-efficient of $x^{-9}$ in $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to ____.

*Space for your notes:*

**Q10 - 29 January - Shift 1**

Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of these three terms, is______.

*Space for your notes:*

**Q11 - 29 January - Shift 2**

Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a}=\frac{2^{\ell}m}{n}$, where m and n are odd numbers, then the ordered pair $(\ell, n)$ is equal to :

(1) (50, 51)
(2) (51, 99)
(3) (50, 101)
(4) (51, 101)

*Space for your notes:*

**Q12 - 30 January - Shift 1**

If the coefficient of $x^{15}$ in the expansion of $\left(ax^3+\frac{1}{bx^{\frac{1}{3}}}\right)^{15}$ is equal to the coefficient of $x^{-15}$ in the expansion of $\left(ax^{\frac{1}{3}}-\frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

(1) a = b (2) ab = 1
(3) a = 3b (4) ab = 3

*Space for your notes:*

**Q13 - 30 January - Shift 1**

The coefficient of $x^{301}$ in $(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+.....+x^{500}$ is:

(1) ${}^{501}C_{302}$ (2) ${}^{500}C_{301}$

(3) ${}^{500}C_{300}$ (4) ${}^{501}C_{200}$

*Space for your notes:*

**Q14 - 30 January - Shift 2**

Let $x=\left(8\sqrt{3}+13\right)^{13}$ and $y=\left(7\sqrt{2}+9\right)^{9}$. If [t] denotes the greatest integer $\le t$, then

(1) [x] + [y] is even
(2) [x] is odd but [y] is even
(3) [x] is even but [y] is odd
(4) [x] and [y] are both odd

*Space for your notes:*

**Q15 - 30 January - Shift 2**

$50^{th}$ root of a number x is 12 and $50^{th}$ root of another number y is 18. Then the remainder obtained on dividing (x + y) by 25 is _____.

*Space for your notes:*

**Q16 - 31 January - Shift 1**

Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}}+\frac{2}{x^3}\right)^{30}$ has a term $\beta x^{-\alpha}, \beta \in \mathbb{N}$.

Then $\alpha$ is equal to __________.

*Space for your notes:*

**Q17 - 31 January - Shift 1**

The remainder on dividing $5^{99}$ by 11 is _______

*Space for your notes:*

**Q18 - 31 January - Shift 2**

The Coefficient of $x^{-6}$, in the expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is ________

*Space for your notes:*

**Q19 - 31 January - Shift 2**

If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ is $-84$ and the Coefficient of $x^{-3\ell}$ is $2^{\alpha}\beta$, where $\beta<0$ is an odd number, Then $|\alpha\ell-\beta|$ is equal to ________

*Space for your notes:*

**Q20 - 01 February - Shift 1**

The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+....+\frac{1}{49!2!}+\frac{1}{51!1!}$ is

(1) $\frac{2^{50}}{50!}$ (2) $\frac{2^{50}}{51!}$

(3) $\frac{2^{51}}{51!}$ (4) $\frac{2^{51}}{50!}$

*Space for your notes:*

**Q21 - 01 February - Shift 1**

The remainder when $19^{200}+23^{200}$ is divided by 49, is _____ .

*Space for your notes:*

**Q22 - 01 February - Shift 2**

Sum of the squares of all possible values of x = 4

If the term without x in the expansion of $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to ______.

Space for your notes:

**Q23 - 01 February - Shift 2**

Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2}\left(10-3^x\right)}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is ______.

Space for your notes:

# Answer Key

**(As per Official NTA Key released on 2 Feb)**

| | | | |
|---|---|---|---|
| **Q1** (1) | **Q2** (1012) | **Q3** (3) | **Q4** (405) |
| **Q5** (2) | **Q6** (1080) | **Q7** (3) | **Q8** (7) |
| **Q9** (1) | **Q10** (1120) | **Q11** (3) | **Q12** (2) |
| **Q13** (4) | **Q14** (1) | **Q15** (23) | **Q16** (2) |
| **Q17** (9) | **Q18** (5040) | **Q19** (98) | **Q20** (2) |
| **Q21** (29) | **Q22** (1) | **Q23** (4) | |

Q1 (1)

$$\sum_{r=0}^{22} {}^{22}C_r \cdot {}^{23}C_r = \sum_{r=0}^{22} {}^{22}C_r \cdot {}^{23}C_{23-r}$$

$$= {}^{45}C_{23}$$

Q2 (1012)

using result

$$\sum_{r=0}^{n} r^2\, {}^nC_r = n(n+1).2^{n-2}$$

$$\text{Then } \sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times 2024 \times 2^{2021}$$

$$= 2023 \times \alpha \times 2^{2022} \text{ So,}$$

$$\Rightarrow \alpha = 1012$$

Q3 (3)

$S=0.({}^{30}C_0)^2+1.({}^{30}C_1)^2+2.({}^{30}C_2)^2+\ldots\ldots+30.({}^{30}C_{30})^2$

$S=30.({}^{30}C_0)^2+29.({}^{30}C_1)^2+28.({}^{30}C_2)^2+\ldots\ldots+0.({}^{30}C_0)^2$

$2S=30.({}^{30}C_0{}^2+{}^{30}C_1{}^2+\ldots\ldots+{}^{30}C_{30}{}^2)$

$S=15\cdot{}^{60}C_{30}=15\cdot\frac{60!}{(30!)^2}$

$\frac{15\cdot10!}{(30!)^2}=\frac{\alpha\cdot60!}{(30!)^2}$

$\Rightarrow\alpha=15$

**Q4 (405)**

Given Binomial $\left(x-\frac{3}{x^2}\right)^n, x \neq 0, n \in N,$

Sum of coefficients of first three terms

$${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 3^2 = 376$$

$$\Rightarrow 3n^2 - 5n - 250 = 0$$

$$\Rightarrow (n-10)(3n+25) = 0$$

$$\Rightarrow n = 10$$

Now general term ${}^{10}C_r x^{10-r}\left(\frac{-3}{x^2}\right)^r$

$$= {}^{10}C_r x^{10-r}(-3)^r \cdot x^{-2r}$$

$$= {}^{10}C_r (-3)^r \cdot x^{10-3r}$$

Coefficient of $x^4 \Rightarrow 10 - 3r = 4$

$$\Rightarrow r = 2$$

$${}^{10}C_2(-3)^2 = 405$$

**Q5 (2)**

$$a_r = {}^{10}C_{10-r} = {}^{10}C_r$$

$$\Rightarrow \sum_{r=1}^{10} r^3\left(\frac{{}^{10}C_r}{{}^{10}C_{r-1}}\right)^2 = \sum_{r=1}^{10} r^3\left(\frac{11-r}{r}\right)^2 = \sum_{r=1}^{10} r(11-r)^2$$

$$= \sum_{r=1}^{10}(121r + r^3 - 22r^2) = 1210$$

**Q6 (1080)**

General term is $\sum \frac{5!(2x)^{n_1}(x^{-7})^{n_2}(3x^2)^{n_3}}{n_1!\,n_2!\,n_3!}$

For constant term,

$n_1 + 2n_3 = 7n_2$

& $n_1 + n_2 + n_3 = 5$

Only possibility $n_1 = 1,\ n_2 = 1,\ n_3 = 3$

$\Rightarrow$ constant term = 1080

**Q7 (3)**

$$\sum_{k=0}^{6} {}^{51-k}C_3$$

$$= {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + \ldots + {}^{45}C_3$$

$$= {}^{45}C_3 + {}^{46}C_3 + \ldots\ldots + {}^{51}C_3$$

$$= {}^{45}C_4 + {}^{45}C_3 + {}^{46}C_3 + \ldots\ldots + {}^{51}C_3 - {}^{45}C_4$$

$$({}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r)$$

$$= {}^{52}C_4 - {}^{45}C_4$$

**Q8 (7)**

$(2023)^{2023}$

$= (2030 - 7)^{2023}$

$= (35\text{ K} - 7)^{2023}$

$= {}^{2023}C_0(35\text{ K})^{2023}(-7)^0 + {}^{2023}C_1(35\text{ K})^{2022}(-7) + \ldots\ldots +$
$\ldots\ldots + {}^{2023}C_{2023}(-7)^{2023}$

$= 35\text{ N} - 7^{2023}.$

Now, $-7^{2023} = -7 \times 7^{2022} = -7\,(7^2)^{1011}$

$= -7\,(50 - 1)^{1011}$

$= -7({}^{1011}C_0\, 50^{1011} - {}^{1011}C_1(50)^{1010} + \ldots\ldots\ {}^{1011}C_{1011})$

$= -7\,(5\lambda - 1)$

$= -35\lambda + 7$

$\therefore$ when $(2023)^{2023}$ is divided by 35 remainder is 7

**Q9 (1)**

Coefficient of $x^9$ in $\left(\alpha x^3 + \frac{1}{\beta x}\right) = {}^{11}C_6 \cdot \frac{\alpha^5}{\beta^6}$

$\because$ Both are equal

$$\therefore {}^{11}C_6 \cdot \frac{\alpha^5}{\beta^6} = -{}^{11}C_5 \cdot \frac{\alpha^6}{\beta^5}$$

$$\Rightarrow \frac{1}{\beta} = -\alpha$$

$$\Rightarrow \alpha\beta = -1$$

$$\Rightarrow (\alpha\beta)^2 = 1$$

**Q10 (1120)**

$$t_{r+1} = {}^nC_r(2x)^r$$

$$\Rightarrow \frac{{}^nC_{r-1}(2)^{r-1}}{{}^nC_r(2)^r} = \frac{2}{5}$$

$$\Rightarrow \frac{\frac{n!}{(r-1)!(n-r+1)!}}{\frac{n!(2)}{r!(n-r)!}} = \frac{2}{5}$$

$$\Rightarrow \frac{r}{n-r+1} = \frac{4}{5} \Rightarrow 5r = 4n - 4r + 4$$

$$\Rightarrow 9r = 4(n+1) \quad \ldots (1)$$

$$\Rightarrow \frac{{}^nC_r(2)^r}{{}^nC_{r+1}(2)^{r+1}} = \frac{5}{8}$$

$$\Rightarrow \frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r+1)!(n-r-1)!}} = \frac{5}{4} \Rightarrow \frac{r+1}{n-r} = \frac{5}{4}$$

$$\Rightarrow 4r + 4 = 5n - 5r \Rightarrow 5n - 4 = 9r \quad \ldots (2)$$

From (1) and (2)

$\Rightarrow 4n + 4 = 5n - 4 \Rightarrow n = 8$

(1) $\Rightarrow r = 4$

so, coefficient of middle term is

**Q11 (3)**

In the expansion of

$(1+x)^{99} = C_0 + C_1x + C_2x^2 + .... + C_{99}x^{99}$

$K = C_1 + C_3 + .... + C_{99} = 2^{98}$

$a \Rightarrow$ Middle in the expansion of $\left(2 + \frac{1}{\sqrt{2}}\right)^{200}$

$$T_{\frac{200}{2}+1} = {}^{200}C_{100}\,(2)^{100}\left(\frac{1}{\sqrt{2}}\right)^{100}$$

$$= {}^{200}C_{100}\,.2^{50}$$

So, $\dfrac{{}^{200}C_{99} \times 2^{98}}{{}^{200}C_{100} \times 2^{50}} = \dfrac{100}{101} \times 2^{48}$

So, $\dfrac{25}{101} \times 2^{50} = \dfrac{m}{n} 2^{\ell}$

$\therefore$ m, n are odd so

$(\ell, n)$ become (50, 101) Ans.

**Q12 (2)**

Option (2)

Coefficient Of $x^{15}$ in $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$

$$T_{r+1}={}^{15}C_r\left(ax^3\right)^{15-r}\left(\frac{1}{bx^{1/3}}\right)^r$$

$$45-3r-\frac{r}{3}=15$$

$$30=\frac{10r}{3}$$

$$r=9$$

Coefficient of $x^{15}={}^{15}C_9\,a^6\,b^{-9}$

Coefficient of $x^{-15}$ in $\left(ax^{1/3}-\frac{1}{bx^3}\right)^{15}$

$$T_{r+1}={}^{15}C_r\left(ax^{1/3}\right)^{15-r}\left(-\frac{1}{bx^3}\right)^r$$

$$5-\frac{r}{3}-3r=-15$$

$$\frac{10r}{3}=20$$

$$r=6$$

Coefficient $={}^{15}C_6\,a^9\times b^{-6}$

$$\Rightarrow \frac{a^9}{b^6}=\frac{a^6}{b^9}$$

**Q13 (4)**

$$(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+\ldots+x^{500}$$

$$=(1+x)^{500}\cdot\left\{\frac{1-\left(\frac{x}{1+x}\right)^{501}}{1-\frac{x}{1+x}}\right\}$$

$$=(1+x)^{500}\frac{\left((1+x)^{501}-x^{501}\right)}{(1+x)^{501}}\cdot(1+x)$$

$$=(1+x)^{501}-x^{501}$$

Coefficient of $x^{301}$ in $(1+x)^{501}-x^{501}$ is given by

$${}^{501}C_{301}={}^{501}C_{200}$$

**Q14 (1)**

$$x = \left(8\sqrt{3}+13\right) = {}^{13}C_0\cdot\left(8\sqrt{3}\right)^{13} + {}^{13}C_1\left(8\sqrt{3}\right)^{12}(13)^1 + \ldots$$

$$x' = \left(8\sqrt{3}-13\right)^{13} = {}^{13}C_0\left(8\sqrt{3}\right)^{13} - {}^{13}C_1\left(8\sqrt{3}\right)^{12}(13)^1 + \ldots$$

$$x - x' = 2\left[{}^{13}C_1\cdot\left(8\sqrt{3}\right)^{12}(13)^1 + {}^{13}C_3\left(8\sqrt{3}\right)^{10}\cdot(13)^3 \ldots\right]$$

therefore, $x - x'$ is even integer, hence [x] is even

Now, $y = \left(7\sqrt{2}+9\right)^9 = {}^9C_0\left(7\sqrt{2}\right)^9 + {}^9C_1\left(7\sqrt{2}\right)^8(9)^1$

$$+{}^9C_2\left(7\sqrt{2}\right)^7(9)^2 \ldots\ldots$$

$$y' = \left(7\sqrt{2}-9\right)^9 = {}^9C_0\left(7\sqrt{2}\right)^9 - {}^9C_1\left(7\sqrt{2}\right)^8(9)^1$$

$$+{}^9C_2\left(7\sqrt{2}\right)^7(9)^2 \ldots\ldots$$

$$y - y' = 2\left[{}^9C_1\left(7\sqrt{2}\right)^8(9)^1 + {}^9C_3\left(7\sqrt{2}\right)^6(9)^3 + \ldots\right]$$

$y - y' =$ Even integer, hence [y] is even

**Q15 (23)**

$$x + y = 12^{50} + 18^{50} = (150-6)^{25} + (325-1)^{25}$$

$$= 25\,K - (6^{25}+1) = 25\,K - ((5+1)^{25}+1)$$

$= 25K_1 - 2$ Remainder $= 23$

**Q16 (2)**

$$T_{r+1} = {}^{30}C_r\left(x^{2/3}\right)^{30-r}\left(\frac{2}{x^3}\right)^r$$

$$= {}^{30}C_r.2^r.x^{\frac{60-11r}{3}}$$

$$\frac{60-11r}{3} < 0 \Rightarrow 11r > 60 \Rightarrow r > \frac{60}{11} \Rightarrow r = 6$$

$$T_7 = {}^{30}C_6.2^6\,x^{-2}$$

We have also observed $\beta = {}^{30}C_6\ (2)^6$ is a natural number.

$\therefore\ \alpha = 2$

**Q17 (9)**

$$5^{99} = 5^4 . 5^{95}$$

$$= 625[5^5]^{19}$$

$$= 625[3125]^{19}$$

$$= 625[3124+1]^{19}$$

$$= 625[11k \times 19 + 1]$$

$$= 625 \times 11k \times 19 + 625$$

$$= 11\,k_1 + 616 + 9$$

$$= 11(k_2) + 9$$

**Q18 (5040)**

$$\left(\frac{4x}{5} + \frac{5}{2x^2}\right)^9,$$

Now, $T_{r+1} = {}^9C_r . \left(\frac{4x}{5}\right)^{9-r} \left(\frac{5}{2x^2}\right)^r$

$$= {}^9C_r . \left(\frac{4}{5}\right)^{9-r} \left(\frac{5}{2}\right)^r . x^{9-3r}$$

Coefficient of $x^{-6}$ i.e. $9\text{-}3r = \text{-}6 \Rightarrow r = 5$

So, Coefficient of $x^{-6} = {}^9C_5 \left(\frac{4}{5}\right)^4 . \left(\frac{5}{2}\right)^5 = 5040$

**Q19 (98)**

In, $\left(\dfrac{x^{\frac{5}{2}}}{2} - \dfrac{4}{x^{\ell}}\right)^9$

$$T_{r+1} = {}^9C_r \frac{\left(x^{5/2}\right)^{9-r}}{2^{9-r}}\left(\frac{-4}{x^{\ell}}\right)^r$$

$$= (-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r x^{\frac{45}{2} - \frac{5r}{2} - lr}$$

$$= 45 - 5r - 2lr = 0$$

$$r = \frac{45}{5 + 2l} \quad \text{------ (1)}$$

Now, according to the question, $(-1)^r \dfrac{{}^9C_r}{2^{9-r}} 4^r = -84$

$$= (-1)^r \, {}^9C_r \, 2^{3r-9} = 21 \times 4$$

Only natural value of r possible if $3r - 9 = 0$

$r = 3$ and ${}^9C_3 = 84$

$\therefore l = 5$ from equation (1)

Now, coefficient of $x^{-31} = x^{\frac{45}{2} - \frac{5r}{2} - lr}$ at $l = 5$, gives

$r = 5$

$$\therefore {}^9c_5 (-1) \frac{4^5}{2^4} = 2^{\alpha} \times \beta$$

$$= -63 \times 2^7$$

$\Rightarrow \alpha = 7, \beta = -63$

$\therefore$ value of $|\alpha\ell - \beta| = 98$

**Q20 (2)**

$$\sum_{r=1}^{26} \frac{1}{(2r-1)!(51-(2r-1))!} = \sum_{r=1}^{26} {}^{51}C_{(2r-1)} \frac{1}{51!}$$

$$= \frac{1}{51!}\left\{{}^{51}C_1 + {}^{51}C_3 + \dots + {}^{51}C_{51}\right\} = \frac{1}{51!}(2^{50})$$

**Q21 (29)**

$(21+2)^{200}+(21-2)^{200}$

$\Rightarrow 2[{}^{100}C_0 21^{200} + 200C_2 21^{198}.\ 2^2 + \ldots.. + {}^{200}C_{198}\ 21^2$

$2^{198} + 2^{200}]$

$\Rightarrow 2[49\ I_1 + 2^{200}] = 49I_1 + 2^{201}$

Now, $2^{201} = (8)^{67} = (1+7)^{67} = 49I_2 + {}^{67}C_0\ {}^{67}C_1.\ 7 =$

$49I_2 + 470 = 49I_2 + 49 \times 9 + 29$

$\therefore$ Remainder is 29

**Q22 (1)**

$$T_{r+1} = {}^{22}C_r.\left(x^{\frac{2}{3}}\right)^{22-r}.(\alpha)^r, x^{-3r}$$

$$= {}^{22}C_r.x^{\frac{44}{3}-\frac{2r}{3}-3r}(\alpha)^r$$

$$\frac{44}{3} = \frac{11r}{3}$$

$r = 4$

$${}^{22}C_4.\alpha^4 = 7315$$

$$\frac{22 \times 21 \times 20 \times 19}{24}.\alpha^4 = 7315$$

$\alpha = 1$

**Q23 (4)**

$$T_6 = {}^mC_5\left(10-3^x\right)^{\frac{m-5}{2}}.\left(3^{x-2}\right) = 21 \quad \ldots.(1)$$

${}^mC_1, {}^mC_2, {}^mC_3$ are in A.P.

$2.\ {}^mC_2 = {}^mC_1 + {}^mC_3$

Solving for m, we get

$m = 2$(rejected), 7

Put in equation (1)

$$21\ .\ (10-3^x)\ \frac{3^x}{9} = 21$$

$3^x = 3^0, 3^2$

$x = 0, 2$